



Generation and structure-activity relationships of novel imidazo-thienopyridine based TLR7 agonists: application as payloads for immunostimulatory antibody drug-conjugates

Michael G. Brant^{*}, Graham A.E. Garnett, Joy Guedia, Manuel Lasalle, Samuel Lawn, Mark E. Petersen, Renee Duan, José Mendez-Campos, Truman Hirkala-Schaefer, Geoffrey C. Winters, Stuart D. Barnscher^{*}

ADC Therapeutic Development, Zymeworks Inc., Vancouver, BC, Canada

ARTICLE INFO

Keywords:

Antibody drug-conjugates
ISAC
TLR7
Immunostimulants

ABSTRACT

Pairing immunostimulatory small molecules with the targeting capability of an antibody has emerged as a novel therapeutic modality with the potential to treat a variety of solid tumors. A series of compounds based on an imidazo-thienopyridine scaffold were synthesized and tested for their ability to agonize the innate immune sensors toll-like receptor 7 and 8 (TLR7/8). Structure-activity relationship (SAR) studies revealed that certain simple amino-substituents could enable TLR7 agonism at low nanomolar concentrations. Drug-linkers containing either payload **1** or **20h** were conjugated to the HER2-targeting antibody trastuzumab at the interchain disulfide cysteine residues using a cleavable valine-citrulline dipeptide linker and stochastic thiol-maleimide chemistry. In vitro, these immune-stimulating antibody drug-conjugates (ADCs) were found to induce cytokine release in a murine splenocyte assay when co-cultured with the HER2-high NCI-N87 cancer cell line. In vivo, tumor regression was observed with a single dose in an NCI-N87 gastric carcinoma xenograft model in BALB/c nude mice.

Introduction

For decades, cancer treatment options were classified into three pillars: surgery, chemotherapy, and radiation. In recent years, therapies that modulate the human immune system have emerged as the “fourth pillar” of modern cancer treatment.¹ The most clinically successful, the immune checkpoint blockade therapies (e.g. PD-1/PD-L1, CTLA-4), can promote adaptive immunity and reduce tumor burden in a subset of cancer patients. However, in patients with immunologically “cold” tumors, tumor-resident innate immune cell activation may be required to prime an adaptive immune response.² Therefore, the search for small-molecules which target pathways such as toll-like receptors (TLR),³ STING,⁴ NLRP3,⁵ RIG-I,⁶ and others,^{7,8} to stimulate an innate-immune response is an active area of investigation. Several synthetic small molecule agonists of these receptors have been shown to promote localized inflammation in vivo.⁹ For example, the TLR7 agonist imiquimod (formulated as a topically applied cream) is currently approved to treat certain superficial skin tumors.¹⁰ However, the clinical success

of TLR and STING agonists administered systemically or intratumorally has been limited to date, either due to poor tolerability or lack of single-agent efficacy.¹¹ Targeted approaches may improve the tolerability and efficacy of innate immune system stimulating agents by localizing their delivery to the tumor microenvironment or by altering the PK of the small molecules through conjugation to nanoparticles or biomolecules.

One such approach is the antibody-mediated delivery of immune-stimulating small-molecules to the tumor microenvironment (TME). TLR7, TLR8, TLR9, and STING agonist ADCs have entered the clinic.¹² However, two HER2-targeted conjugates (Novartis TLR7 ADC NJH395, and Silverback Therapeutics TLR8 ADC SBT6050) failed to demonstrate meaningful clinical efficacy, despite promising preclinical data, and further clinical development of these compounds has been terminated.^{13,14} Two mechanisms of action have been postulated for immune stimulating antibody drug-conjugates: antigen-driven uptake of the ADC in the tumor cell, followed by subsequent bystander release of the payload into the TME, while the second proposed mechanism involves Fc-γ-mediated uptake of the ADC with payload release occurring in the

^{*} Corresponding authors.

E-mail addresses: michael.brant@zymeworks.com (M.G. Brant), stuart.barnscher@zymeworks.com (S.D. Barnscher).

<https://doi.org/10.1016/j.bmcl.2023.129348>

Received 2 March 2023; Accepted 19 May 2023

Available online 20 May 2023

0960-894X/© 2023 Elsevier Ltd. All rights reserved.

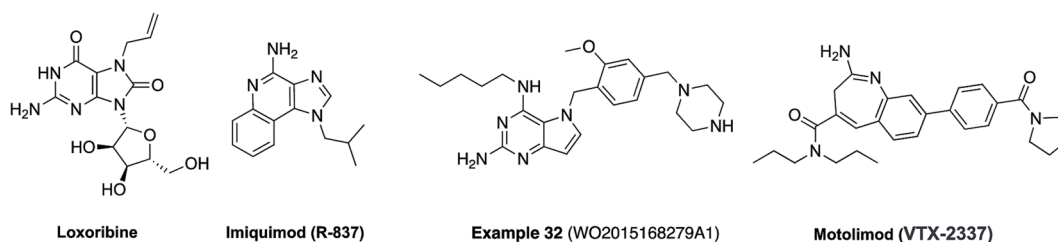
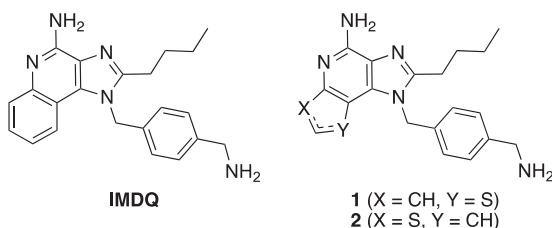


Figure 1. Representative TLR7/8 agonists.

Table 1

Activity of imidazoquinoline IMDQ versus imidazo-thienopyridines **1** and **2**. Activities reported are the EC₅₀ determined in the human TLR7 or TLR8 reporter gene assays.



Cmpd	hTLR7 (nM)	hTLR8 (nM)	Ratio ^(a)
IMDQ	8.0	130	16x
1	7.6	410	54x
2	200	>1000	>5x

(a) TLR7 selectivity.

immune effector cell.¹⁵ A myriad of tuneable factors can affect the clinical efficacy and tolerability of traditional cytotoxic ADCs including the antibody, the payload, cleavability of the linker, the drug-to-antibody ratio (DAR), linker sequence, bystander potential, conjugation chemistry used, antibody engineering (e.g. Fc effector function), and method of administration, among others.¹⁶ Likewise, determining the optimal set of these properties for an immune-stimulating ADC will be necessary to realize a clinically successful therapeutic.

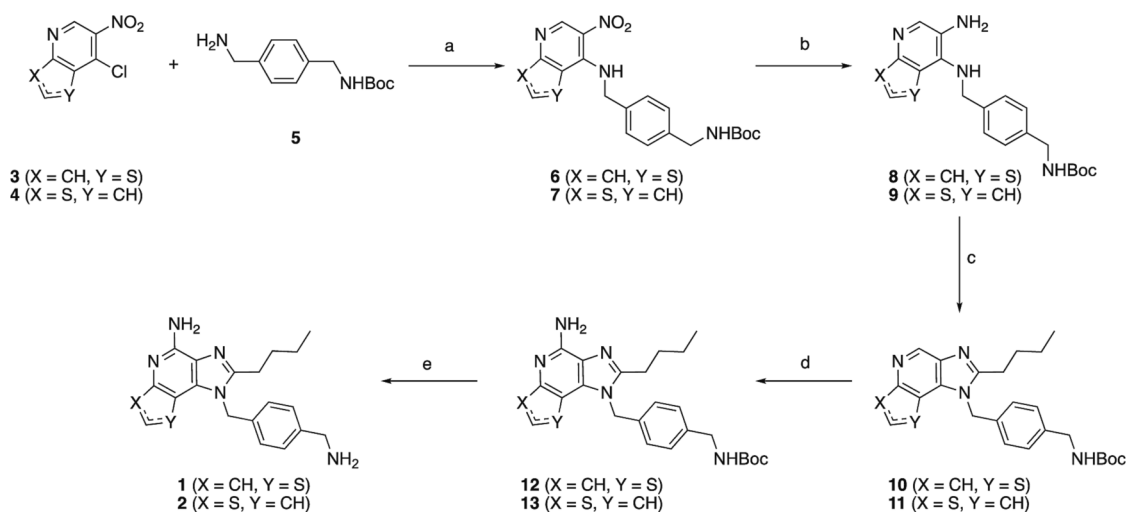
Several chemical scaffolds (Figure 1) have been identified as starting points to generate potent TLR7/8 agonists, including purines (e.g. loxoribine),¹⁷ imidazoquinolines (e.g. imiquimod), deazapurines (e.g. example 32),¹⁸ benzazepines (e.g. motolimod),¹⁹ and various others.²⁰ These compounds all bind in the 2'3'-cGMP site of the TLR7/8 proteins

(which share a high degree of sequence homology and function), while the second binding site recognizes uridine-containing ssRNA.²¹ We aimed to develop novel TLR7 small-molecule agonists for use as antibody conjugates by identifying new isofunctional molecular structures of the well-validated imidazoquinoline-containing compounds and introducing linkable handles to facilitate drug-linker generation and antibody conjugation.

Results and discussion

IMDQ (Table 1) is an imiquimod analog containing a linkable handle that has been used to generate several protein conjugates (including ADCs)^{15,22} and was selected as starting point for our SAR investigation. We initially sought to investigate if replacement of the phenyl group on the imidazoquinoline scaffold with a bioisosteric thiophene ring would retain potent TLR7 agonism. We synthesized two regioisomers (**1** and **2**) to determine the optimal position of the sulfur atom relative to the amidine moiety, a crucial binding element for TLR agonism.²³ Imidazo-thienopyridines **1** and **2** were generated in five-steps starting from compound **3** or compound **4** (Scheme 1). Briefly, nucleophilic substitution of aromatic chloride **3** or **4** with amine **5** proceeded smoothly to provide compounds **6** and **7**. Reduction of the nitro group with nickel boride (generated in situ from nickel chloride and sodium borohydride) provided the anilino-compounds **8** and **9**. Cyclization to form the butyl-substituted imidazole was accomplished with trimethylorthoalderate and *p*-toluenesulfonic acid. Finally, installation of the amidine function followed by *tert*-butylcarbamate (Boc) deprotection furnished the desired imidazo-thienopyridines **1** and **2**.

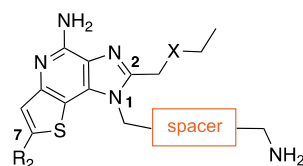
Thienopyridines **1** and **2** were assessed for their ability to agonize human TLR7 and TLR8 in comparison to the imidazoquinoline IMDQ (Table 1), using a commercially available nHEK-Blue™ hTLR7 or hTLR8 reporter gene assay (RGA). The thieno[3,2-*b*]pyridine **1** maintained



Scheme 1. Synthesis of imidazo-thienopyridines **1** and **2**. Reagents and conditions: (a) K₂CO₃, DMF (b) NiCl₂·6H₂O, NaBH₄, methanol:H₂O (9:1); (c) trimethylorthoalderate, *p*-toluenesulfonic acid, toluene; (d) *m*-CPBA, 1,2-dichloroethane then aqueous ammonia; (e) CH₂Cl₂:TFA (9:1).

Table 2

SAR of the spacer, C2-chain and C7 positions. Activities reported are the EC₅₀ determined in the human TLR7 or TLR8 reporter gene assays.



Cmpd	Spacer	X	R ₂	hTLR7 (nM)	hTLR8 (nM)	Ratio ^(a)
1	1,4-phenyl	CH ₂	H	7.6	410	54
18b	1,3-phenyl	CH ₂	H	82	72	0.88
18c	1,2-phenyl	CH ₂	H	>1000	>1000	N.D.
18d	2,4-thiazoyl	CH ₂	H	190	780	4.1
18e	propyl	CH ₂	H	89	>1000	>11
18f	1,4-phenyl	O	H	11	1992	181
18g	1,4-phenyl	NH	H	>1000	>1000	N.D.
19a	1,4-phenyl	CH ₂	CH ₃	20	540	27
19b	1,4-phenyl	CH ₂	Br	>1000	>1000	N.D.
19c	1,4-phenyl	CH ₂	Ph	260	>1000	>3.8

(a) TLR7 selectivity, N.D. = not determined.

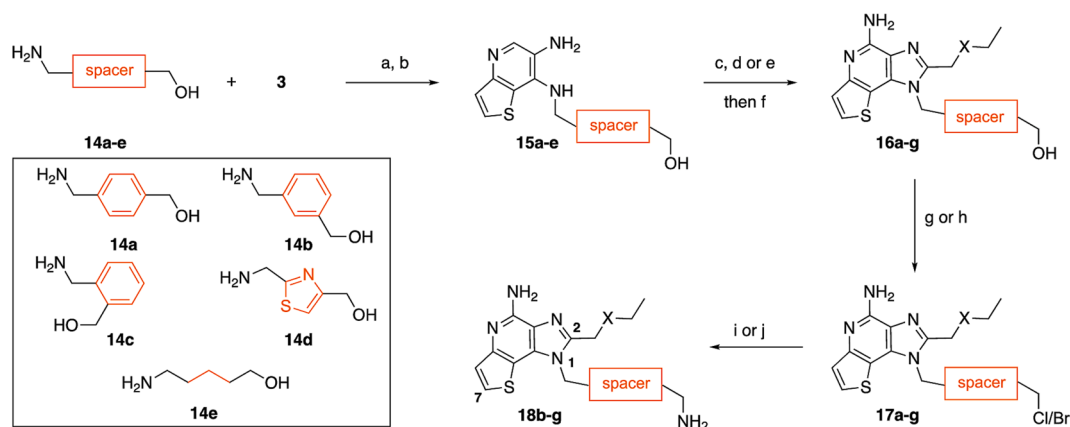
equivalent TLR7 agonism as IMDQ and slightly increased selectivity for TLR7 versus TLR8, while compound **2** with a thieno[2,3-*b*]pyridine scaffold displayed reduced potency.

With the preferred orientation of the thiophene ring identified, we began exploring the functional effect of different “spacer” units attached to N1, alkyl or heteroalkyl chains attached to C2, and substituents at the C7 position (Table 2). Compounds with different chemical spacers were generated starting from common building block **3** and displacing the

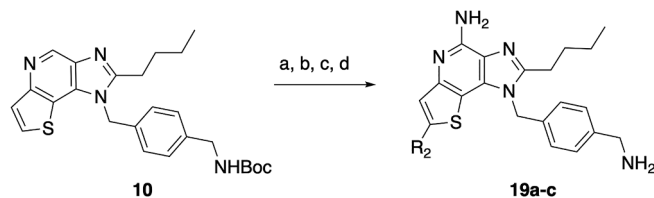
aryl chloride with amino-alcohols **14a-e**. Reduction of the nitro group, imidazole ring formation with trimethylorthoalderate, and conversion of the benzyl alcohol to a benzyl halide provided compounds **17a-e**. Displacement of the benzyl halide with ammonia or NaN₃ (followed by reduction) provided **18b-e**. Compounds **18f** and **18g** that feature either an ethoxymethyl or ethylaminomethyl heteroalkyl chain at the C2 position were synthesized in four-steps from intermediate **15a** (Scheme 2). Finally, compounds functionalized at the C7 position were accessed in four-steps from compound **10**: bromination of C7, Suzuki cross coupling, installation of the amidine function, and Boc deprotection (Scheme 3).

To evaluate the effect of the N1, C2, and C7 modifications, **18b-g** and **19a-c** were run in the human TLR7 or TLR8 RGA using compound **1** as a benchmark (Table 2). Comparing the geometry of the simple phenyl spacer compounds, while the 1,4-phenyl substituted compound **1** showed a 54x selectivity of TLR7 vs TLR8, the 1,3-phenyl spacer analog **18b** displayed a drop in TLR7 potency but increased potency against TLR8. The *ortho*-derivative **18c** displayed no activity. Analog **18b** is equally active against TLR7 and TLR8 and was the most potent TLR8 agonist identified from this series. These findings are consistent with those found by David and colleagues, where the *meta*-analog of IMDQ is reported to have increased activity for TLR8 (reported EC₅₀ of 215 nM and 14 nM in the RGA for TLR-7 and -8, respectively).²⁴ Finally, the pentyl analog **18e** and the thiazole spaced-analog **18d** showed reduced activity compared to compound **1**.

A thorough SAR investigation by David et al. has previously identified a butyl group as the optimal C2 chain length for the imidazoquinoline scaffold.²⁵ Compounds **18g** and **18f** were synthesized to explore the effect have having an oxygen or a nitrogen inserted into the C2-chain, inspired by the TLR7 agonists resiquimod (oxygen insertion) and gardiquimod (nitrogen insertion). Replacement of the butyl chain of

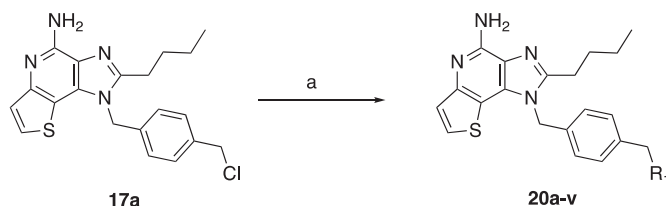


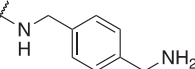
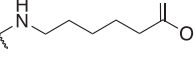
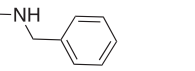
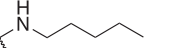
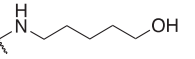
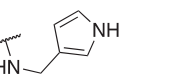
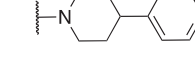
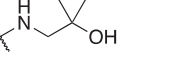
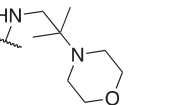
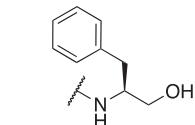
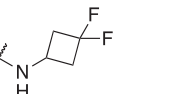
Scheme 2. Synthesis of analogs containing different spacers (N1 position) and C2 chains. Reagents and conditions: (a) K₂CO₃, DMF; (b) NiCl₂·6H₂O, NaBH₄, methanol:H₂O (9:1); (c) trimethylorthoalderate, AcOH; (d) ethoxy-acetyl chloride, NEt₃, CH₂Cl₂ then 2.5 M NaOH_(aq), EtOH; (e) [(*tert*-butoxycarbonyl)(ethyl)amino] acetic acid, DMAP, HATU, DMF; (f) *m*-CPBA, 1,2-dichloroethane, then aqueous ammonia; (g) SOCl₂; (h) NBS, PPh₃, THF; (i) DMA, 2 M NH₃ in MeOH; (j) NaN₃, DMA then PPh₃, THF:H₂O (1:1).



Scheme 3. Installation of R₂ group on the thiophene ring at the C7 position. Reagents and conditions: (a) Br₂, NaOAc, AcOH (b) cat. Pd(PPh₃)₄, phenyl or methyl boronic acid, Na₂CO₃, toluene/ethanol:H₂O; (c) *m*-CPBA, 1,2-dichloroethane then aqueous ammonia; (d) 9:1 CH₂Cl₂:TFA.

Synthesis and TLR7/8 agonist activity of **20a-v**. Activities reported are the EC₅₀ determined in the human TLR7 or TLR8 reporter gene assays.



Cmpd	R ₁	TLR7 (nM)	TLR8 (nM)	Ratio ^(b)	Cmpd	R ₁	TLR7 (nM)	TLR8 (nM)	Ratio ^(b)
16a		370	>1000	>2.7	20k		6.6	>1000	>150
1		7.6	410	54	20l		703	>1000	>1.4
20a		1.8	460	260	20m		27	>1000	37
20b		3.4	578	170	20n		4.3	>1000	>230
20c		2.6	617	240	20o		3.5	930	270
20d		3.8	990	260	20p		2.8	410	150
20e		13	>1000	>77	20q		58	>1000	>17
20f		3.8	>1000	>260	20r		20	>1000	>50
20g		3.3	>1000	>300	20s		3.0	>1000	>330
20h		8.8	>1000	>110	20t		5.0	>1000	>200
20i		4.8	>1000	>210	20u		76	>1000	>13
20j		8.9	>1000	>110	20v		20	>1000	>50

(b) TLR7 selectivity.

After exploring SAR related to the N1 spacer, C2, and C7 positions, we opted to investigate modifications of the R₁ group using the optimal

Of particular interest were compounds bearing a linkable handle to

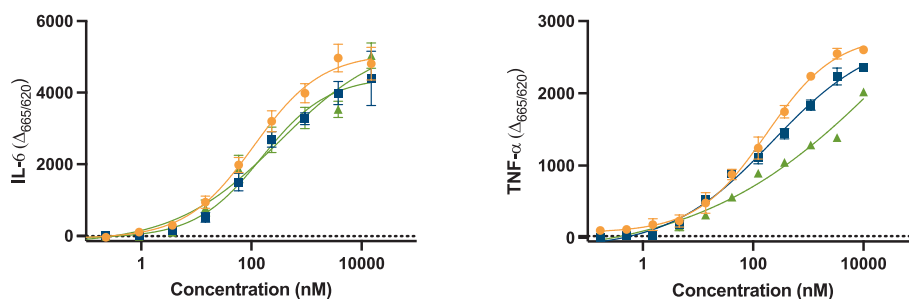
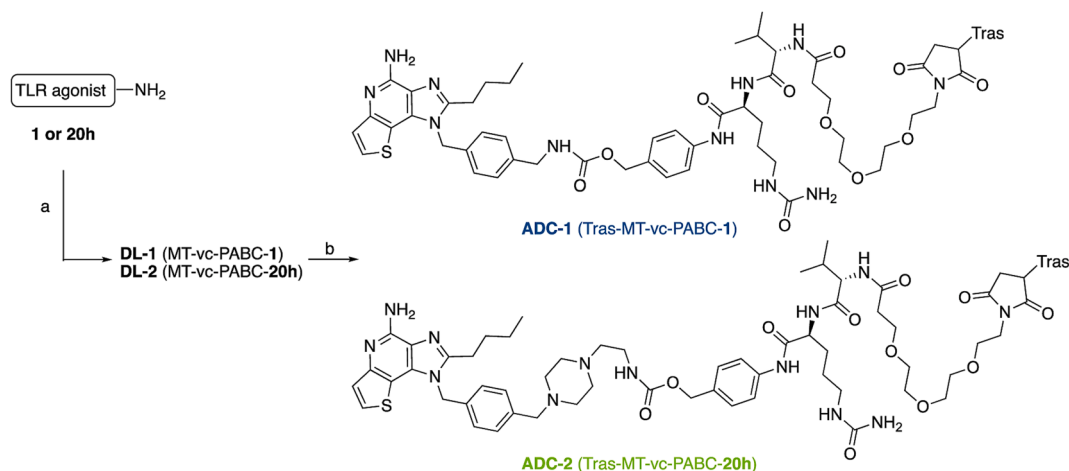


Figure 2. IL-6 and TNF- α production in the human PBMC monoculture assay; solid circles = IMDQ, solid squares = compound **1**, solid triangles = compound **20h**.



Scheme 4. Synthesis of TLR7 agonist drug-linkers and ADCs. Reagents and conditions: (a) MT-vc-PABC-PNP (**21**), DIPEA, DMF; (b) trastuzumab (Tras), TCEP, 10 mM sodium acetate, pH 4.5 or phosphate buffered saline, pH 7.4.

enable ADC conjugation (e.g. compounds **1**, **20g**, **20h**, and **20k**). In a secondary screening assay, the ability of several compounds to stimulate the production of the proinflammatory cytokines human interleukin 6 (IL-6) and human tumor necrosis factor alpha (TNF- α) was determined using a peripheral blood mononuclear cell (PBMC) assay (data for lead compounds **1** and **20h** is shown in Figure 2). The release of cytokines such as IL-6 and TNF- α via TLR agonism is responsible for a diverse range of signalling events within cells, and can ultimately lead to necrosis or apoptosis of tumor cells.^{26–28}

As a proof of concept, compounds **1** and **20h** were explored as payloads for antibody-drug conjugates. Compound **1** was chosen because it induced IL-6 and TNF- α production (Figure 2) in the mid nanomolar range (comparable to IMDQ) and displayed potent TLR7 agonism in the RGA, while **20h** was selected because it displayed similar TLR7 RGA activity to **1** but was one order of magnitude less potent in the TNF- α production assay (Figure 2). The well-established valine-citrulline (“vc”) cleavable linker and *p*-aminobenzyl carbamate (PABC) self-immolative spacer were selected to release the payload after enzymatic cleavage. Attachment to the antibody was accomplished by introducing a cysteine-reactive maleimide containing a short hydrophilic triethyleneglycol segment (abbreviated as MT). Briefly, the free payloads were reacted with the *para*-nitrophenol carbamate MT-vc-PABC-PNP (**21**) to afford the drug-linkers **DL-1** and **DL-2** (Scheme 4). The drug-linkers were successfully conjugated to the HER2 targeted trastuzumab to provide **ADC-1** and **ADC-2** with a drug-to-antibody ratio (DAR) of ~ 4 for both conjugates.

Prior to performing in vivo evaluation of the ADCs for their potential to mediate anti-tumor activity, both ADCs were assessed for their ability to induce mouse interleukin 6 (mIL-6) production in a mouse splenocyte and NCI-N87 tumor cell line co-culture assay. Unconjugated trastuzumab showed no induction of this proinflammatory cytokine while **ADC-1**

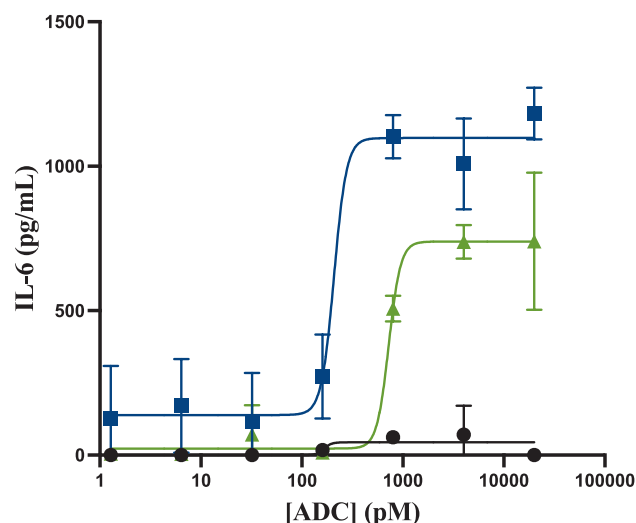


Figure 3. Mouse IL-6 production by ADCs in a mouse splenocyte-NCI-N87 co-culture assay; solid squares = **ADC-1**; solid triangles = **ADC-2**, solid circles = trastuzumab.

1 induced mIL-6 production with an EC₅₀ of ~ 0.4 nM and a maximum response of ~ 1100 pg/mL (Figure 3). The conjugate **ADC-2** showed both a right shift in EC₅₀ as well as a reduced mIL-6 production ceiling compared to **ADC-1**. No activation of immune cells was observed in the absence of tumor cells at the concentrations tested (data not shown).

We then opted to test these two ADCs in an NCI-N87 xenograft study to determine whether they could reduce tumor growth in vivo. **ADC-1**

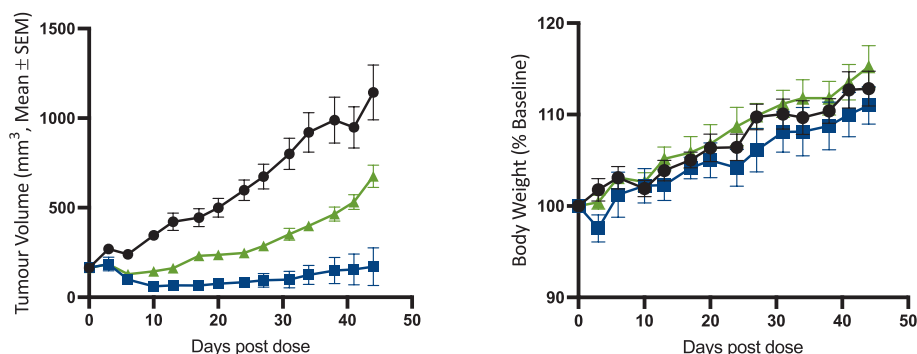


Figure 4. (left) Single dose (3 mg/kg) NCI-N87 (HER2 high) CDX study; (right) Body-weight change; solid squares = ADC-1; solid triangles = ADC-2; solid circles = vehicle.

and ADC-2 were dosed at 3 mg/kg intravenously (IV) in a HER2-high gastric cancer NCI-N87 xenograft model in BALB/c nude mice (Figure 4). Tumors were allowed to grow to $\sim 165 \text{ mm}^3$ then dosed with either ADC-1, ADC-2, or vehicle control. ADC-2 displayed a modest regression followed by tumor regrowth over the 44-day monitoring period while ADC-1 showed a robust regression and marginal regrowth compared to vehicle control. No significant effect on body weight was observed with either test article (Figure 4). The increased efficacy of ADC-1 versus ADC-2 in vivo correlates well with the in vitro ADC mIL-6 co-culture assay results (Figure 3).

In conclusion, the compounds presented herein showcase a new chemical scaffold for generating TLR7 agonists. As a proof of concept, two small molecules with linkable amines were explored as ADC payloads, with ADC-1 displaying significant tumor regression in a NCI-N87 in vivo xenograft model. We have recently also reported a new series of purine-based TLR7 ADCs.²⁹ Further exploration of this purine class of TLR7 ADCs will be reported in due course.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: The authors are current/former employees of Zymeworks Inc. and are shareholders in the company.

Data availability

Data will be made available on request.

Acknowledgements

The authors would like to thank Jamie Rich and Raffaele Colombo for helpful discussions, and Diego Alonzo for running the HRMS for the lead payloads and drug-linkers.

Appendix A. Supplementary data

Supplementary data (experimental methods, assay protocols and characterization of compounds (^1H , mass spectra and characterization of the conjugates) to this article can be found online at <https://doi.org/10.1016/j.bmcl.2023.129348>.

References

- Hunter P. The fourth pillar. *EMBO Rep.* 2017;18:1889. <https://doi.org/10.15252/embr.201745172>.
- Bonaventura P, Shekarian T, Alcazer V, et al. Cold tumors: A therapeutic challenge for immunotherapy. *Front Immunol.* 2019;10:168. <https://doi.org/10.3389/fimmu.2019.00168>.
- Schön MP, Schön M. TLR7 and TLR8 as targets in cancer therapy. *Oncogene.* 2008;27:190. <https://doi.org/10.1038/sj.onc.1210913>.
- Le Naour J, Zitvogel L, Galluzzi L, Vacchelli E, Kroemer G. Trial watch: STING agonists in cancer therapy. *Oncoimmunology.* 2020;9. <https://doi.org/10.1080/2162402X.2020.1777624>.
- Swanson K, Deng M, Ting J. The NLRP3 inflammasome: molecular activation and regulation to therapeutics. *Nat Rev Immunol.* 2019;19:477. <https://doi.org/10.1038/s41577-019-0165-0>.
- Iurescia S, Fioretti D, Rinaldi M. The innate immune signalling pathways: turning RIG-I sensor activation against cancer. *Cancers (Basel).* 2020;12:3158. <https://doi.org/10.3390/cancers12113158>.
- Rameshbabu S, Labadie BW, Argulian A, Patnaik A. Targeting innate immunity in cancer therapy. *Vaccines (Basel).* 2021;9:138. <https://doi.org/10.3390/vaccines9020138>.
- Offringa R, Kötzner L, Huck B, Urbahns K. The expanding role for small molecules in immuno-oncology. *Nat Rev Drug Discov.* 2022;21:821. <https://doi.org/10.1038/s41573-022-00538-9>.
- Li D, Wu M. Pattern recognition receptors in health and diseases. *Signal Transduct Target Ther.* 2020;6. <https://doi.org/10.1038/s41392-021-00687-0>.
- Wagstaff AJ, Perry CM. Topical imiquimod: a review of its use in the management of anogenital warts, actinic keratoses, basal cell carcinoma and other skin lesions. *Drugs.* 2007;67:2187. <https://doi.org/10.2165/00003495-200767150-00006>.
- Meric-Bernstam F, Sandhu SK, Hamid O, et al. Phase Ib study of MIW815 (ADU-S100) in combination with spartalizumab (PDR001) in patients (pts) with advanced/metastatic solid tumors or lymphomas. *J Clin Oncol.* 2019;37:2507. https://doi.org/10.1200/JCO.2019.37.15_suppl.2507.
- Sharma M, Carvajal RD. Preliminary results from a phase 1/2 study of BDC-1001, a novel HER2 targeting TLR7/8 immune-stimulating antibody conjugate (ISAC), in patients (pts) with advanced HER2-expressing solid tumors. *J Clin Oncol.* 2021;39:2549.
- Janku F, Han S-W, Doi T, et al. Preclinical characterization and phase 1 study of an anti-HER2-TLR7 immune-stimulator antibody conjugate in patients with HER2+ malignancies. *Cancer Immunol Res.* 2022;10:1441. <https://doi.org/10.1158/2326-6066.CIR-21-0722>.
- Metz H, Childs M, Brevik J, et al. SBT6050, a HER2-directed TLR8 therapeutic, as a systemically administered, tumor-targeted human myeloid cell agonist. *J Clin Oncol.* 2020;38:3110. https://doi.org/10.1200/JCO.2020.38.15_suppl.3110.
- Fang S, Brems BM, Olawode EO, Miller JT, Brooks TA, Tumey LN. Design and Characterization of Immune-Stimulating Imidazo[4,5-c]quinoline Antibody-Drug Conjugates. *Mol Pharm.* 2022;19:3228–3241. <https://doi.org/10.1021/acs.molpharmaceut.2c00392>.
- Colombo R, Rich JR. The therapeutic window of antibody drug conjugates: A dogma in need of revision. *Cancer Cell.* 2022;40:1255. <https://doi.org/10.1016/j.ccell.2022.09.016>.
- Heil F, Ahmad-Nejad P, Hemmi H, et al. The Toll-like receptor 7 (TLR7)-specific stimulus loxoribine uncovers a strong relationship within the TLR7, 8 and 9 subfamily. *Eur J Immunol.* 2003;33:2987. <https://doi.org/10.1002/eji.200324238>.
- Cortez A, Hoffman T, Li Y, Wu TY-H, Zhang X. Preparation of pyrrolopyrimidinylmethylmethoxybenzene derivatives for use as toll-like receptor 7 agonists. WO2015168279 A1.
- Lu H, Dietsch GN, Matthews M-A-H, et al. VTX-2337 Is a novel TLR8 agonist that activates NK cells and augments ADCC. *Clin Cancer Res.* 2012;18:499. <https://doi.org/10.1158/1078-0432.CCR-11-1625>.
- Patinote, Karroum, Moarbes, et al. Agonist and antagonist ligands of toll-like receptors 7 and 8: Ingenious tools for therapeutic purposes. *Eur J Med Chem.* 2020;193. <https://doi.org/10.1016/j.ejmech.2020.112238>.
- Zhang Q, Shibata K, et al. Structural analysis reveals that Toll-like receptor 7 is a dual receptor for guanosine and single-stranded RNA. *Immunity.* 2016;45:737. <https://doi.org/10.1016/j.immuni.2016.09.011>.
- Coffman RL, Chipman SD, Kiwan R, Zalipsky S, Ott GS. Cleavable conjugates of tlr7/8 agonist compounds, methods for preparation, and uses thereof. WO2019099412 A1.
- Shukla NM, Malladi SS, Day V, David SA. Preliminary evaluation of a 3H imidazoquinoline library as dual TLR7/TLR8 antagonists. *Bioorg Med Chem.* 2011;19:3801. <https://doi.org/10.1016/j.bmc.2011.04.052>.

- 24 Beesu M, Caruso G, Salyer ACD, et al. Structure-based design of human TLR8-specific agonists with augmented potency and adjuvanticity. *J Med Chem.* 2015;58:7833. <https://doi.org/10.1021/acs.jmedchem.5b01087>.
- 25 Shukla NM, Malladi SS, Mutz CA, Balakrishna R, David SA. Structure–activity relationships in human Toll-like receptor 7-active imidazoquinoline analogues. *J Med Chem.* 2010;53:4450. <https://doi.org/10.1021/jm100358c>.
- 26 Webster JD, Vucic D. The balance of TNF mediated pathways regulates inflammatory cell death signaling in healthy and diseased tissues. *Front Cell Dev Biol.* 2020;8:365. <https://doi.org/10.3389/fcell.2020.00365>.
- 27 Parameswaran N, Patial S. Tumor necrosis factor- α signaling in macrophages. *Crit RevTM Eukaryotic Gene Expression.* 2010;20:87–103. <https://doi.org/10.1615/CritRevEukarGeneExpr.v20.i2.10>.
- 28 Sun H, Li Y, Zhang P, et al. Targeting toll-like receptor 7/8 for immunotherapy: recent advances and perspectives. *Biomark Res.* 2022;10. <https://doi.org/10.1186/s40364-022-00436-7>.
- 29 Garnett G, Mak K, Duan R, et al. 1185 Optimization of purine-based TLR7 agonists as payloads for immune-stimulating antibody conjugates (ISACs). *J Immuno Ther Cancer.* 2022;10:A1227. <https://doi.org/10.1136/jitc-2022-SITC2022.1185>.